



LATENT-SPACE DENSITY MODELING FOR RESOLVABLE-SOURCE DETECTION IN CONFUSION-LIMITED LISA SIMULATIONS



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INTRODUCTION

LISA will operate in a confusion-limited regime where unresolved Galactic binaries form a structured foreground across much of the millihertz band [1]. In this setting, the problem is not simply detecting weak signals in instrumental noise, but distinguishing resolvable sources embedded within overlapping background structure. Here we study one-class detection using continuous wavelet transform (CWT) representations and latent embeddings learned by a convolutional autoencoder trained only on background segments [2]. We compare local geometric deviation in latent space against explicit latent density modeling under a fixed preprocessing and data-generation pipeline.

DATA GENERATION

Confusion background = Instrumental noise + unresolved Galactic binaries

Positive samples = Background + injected resolvable source (MBHB, EMRI, bright GB)

Segment duration = 3600 s sampled at 1 Hz

Training/Test set = 5000 background segments / 200 background + 400 signal segments

Representation = CWT scalograms

SCORING METHODS

$$\text{AE+Manifold}^* \quad \alpha e_{\text{AE}} + \beta d_{\mathcal{M}}$$

$$\text{AE+Manifold+Morph}^* \quad (1 - \lambda)s + \lambda d_{\phi}$$

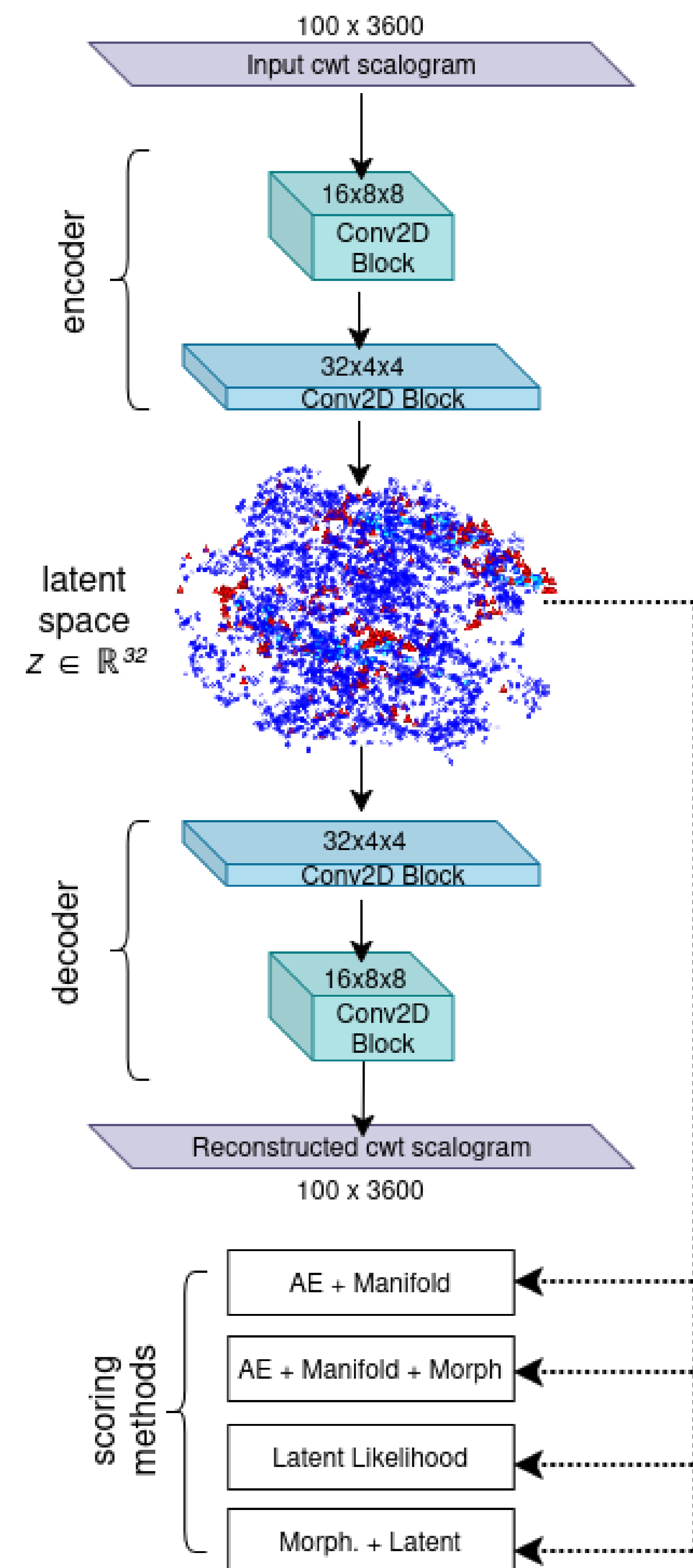
$$\text{Latent Likelihood}^{**} \quad -\log p(f_{\theta}(x))$$

$$\text{Morph+Latent}^{**} \quad -\log p((\phi(x), f_{\theta}(x)))$$

*Local latent deviation.

**GMM or KDE estimators.

LATENT SPACE FRAMEWORK



PRIMARY RESULTS

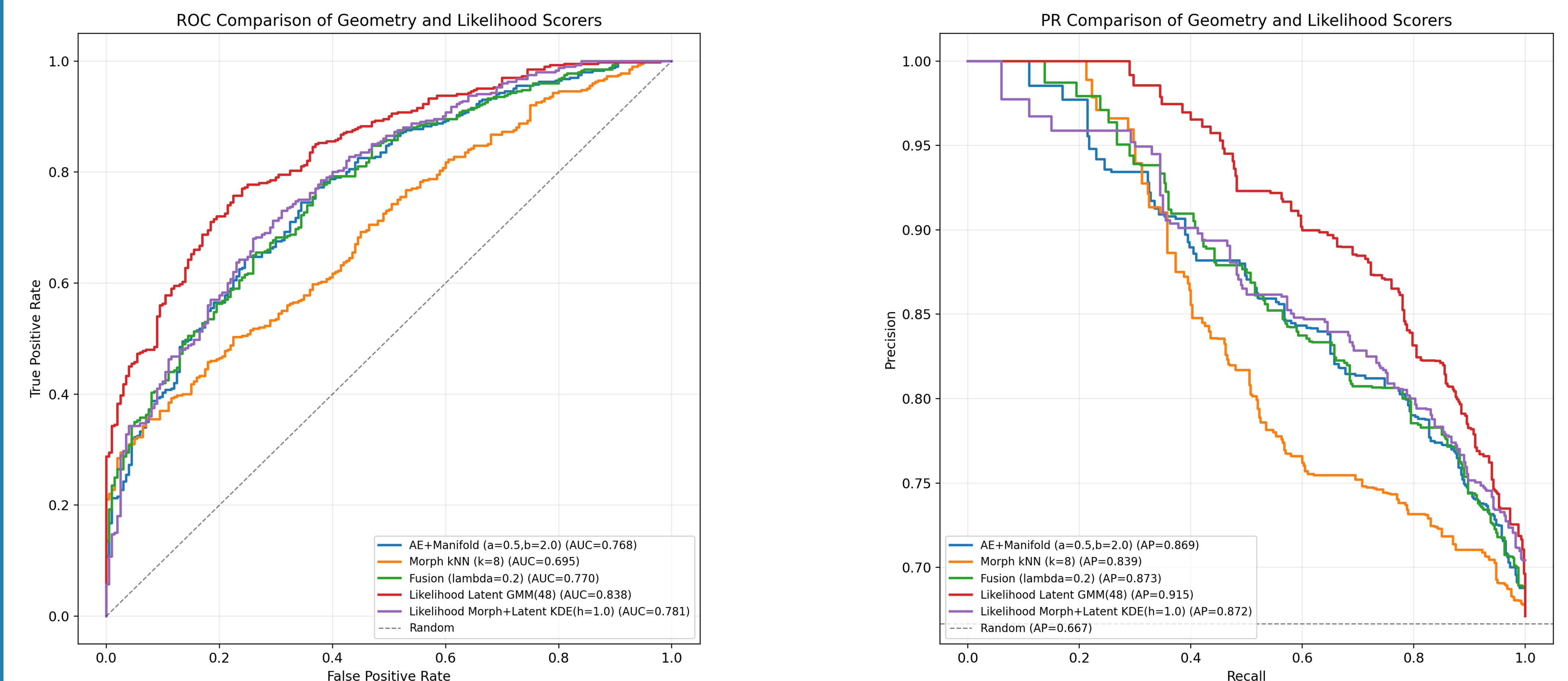


Figure 1: Representative single-run ranking curves across methods. Left: ROC (TPR vs FPR). Right: precision-recall. Legend values correspond to threshold free summary scores for that run [3].

HYPERPARAMETER ABLATION

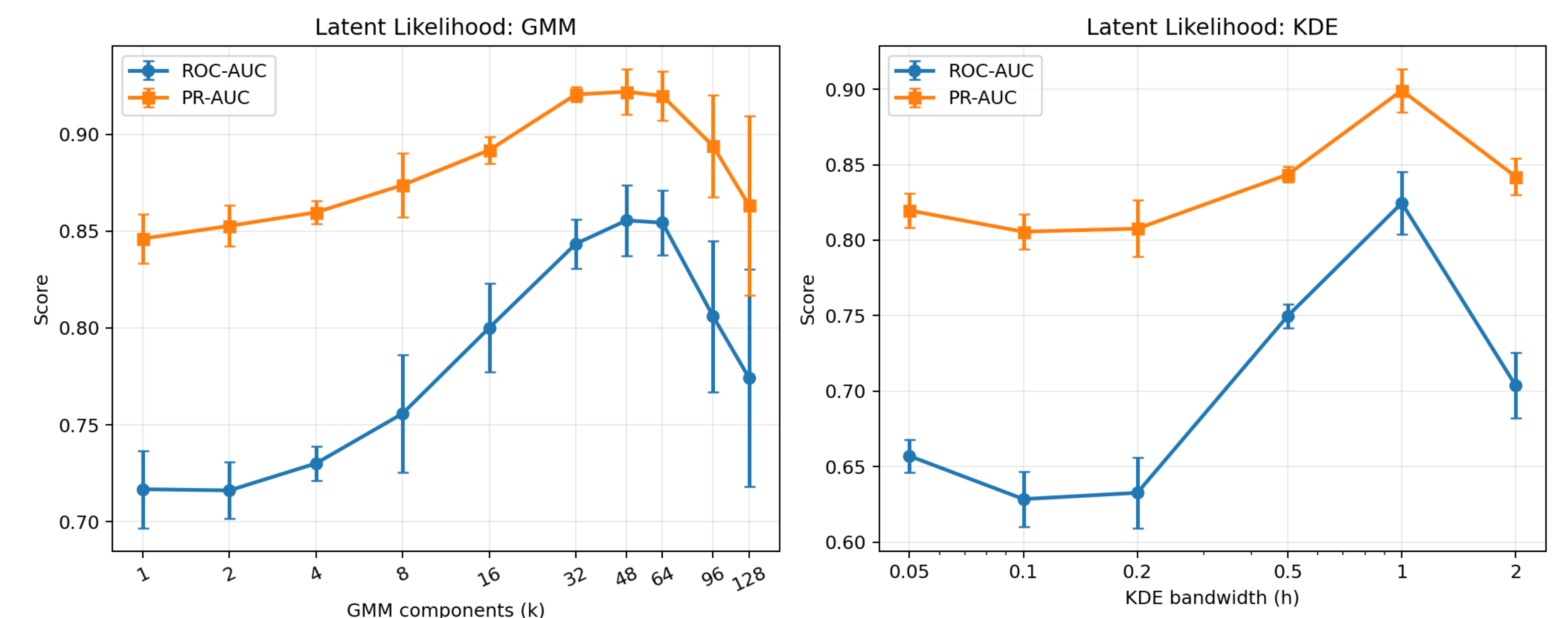


Figure 2: Latent-likelihood ablation. Left: ROC-AUC and PR-AUC versus GMM component count. Right: ROC-AUC and PR-AUC versus KDE bandwidth (log-scaled). Values are mean $\pm$ std across three seeds.

REFERENCES

- [1] LISA Collaboration. Laser interferometer space antenna, 2017. arXiv:1702.00786.
- [2] Jericho Cain. Manifold learning for source separation in confusion-limited gravitational-wave data, 2025. arXiv:2511.12845.
- [3] Jericho Cain. Likelihood-based one-class scoring in cwt latent space for confusion-limited lisa gravitational-wave detection, 2026. arXiv:2602.20212.

DISCUSSION

Likelihood-based scoring consistently outperforms local manifold-distance methods under the fixed latent representation studied here, suggesting that the confusion background contains global distributional structure not fully captured by nearest-neighbor geometry alone. Future work will examine whether these scoring methods remain stable under invertible transformations of the latent coordinates, including rotations, whitening, and nonlinear reparameterizations of the latent space.